

Optimizing a coarse-grained model for large-scale membrane protein simulation

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ABSTRACT Coarse-grained (CG) models are widely used to study membrane proteins at physiologically relevant scales. However, simulating long-range bilayer deformations induced by membrane-embedded proteins at submicrometer scales remains challenging. Here, we assess a generic solvent-free CG lipid model, previously applied to membrane proteins, for large-scale molecular dynamics simulations. We find that beyond a critical membrane size, the model becomes unstable due to membrane poration and unphysical undulations. To overcome this limitation, we systematically optimize this lipid model, significantly extending its stability for larger membrane systems. Using this improved model, we simulate membrane deformation induced by the mechanosensitive ion channel PIEZO in bilayers with varying mechanical properties. This optimized CG model with tunable mechanical properties provides a timely tool for investigating bilayer-mediated membrane protein interactions and bridging the gap between continuum elasticity theory and atomistic simulations.

WHY IT MATTERS Cell membranes are dynamic environments where proteins sense, remodel, and interact through the bilayer. Capturing such phenomena at biologically relevant scales is possible with both experimental and computational approaches. On the computational side, coarse-grained lipid models provide a useful framework for simulating mesoscopic systems. We describe how the Grime lipid model can systematically be adapted to overcome scale-dependent instabilities, enabling robust simulation of membrane and protein-membrane systems: using PIEZO as the test case, we demonstrate how this approach captures its membrane footprint at submicrometer scales.

INTRODUCTION

Cell membranes deform in response to mechanical stresses (1). The creation and detection of membrane deformation by membrane-embedded proteins are critical for understanding the physical environment surrounding cells and for transducing mechanical stimuli into electrochemical signals (2). Computational simulation techniques, such as molecular dynamics (MD), have become increasingly useful for studying membrane proteins due to their high spatial and temporal resolution (3–5). However, simulating cellular processes involving membrane proteins at micrometer scales using MD at atomic resolution (e.g., CHARMM (6), AMBER models (7),

or united atom models (8,9)) or near-atomic resolution (e.g., Martini models (10)) remains challenging. These cellular processes occur on timescales that are orders of magnitude longer than the typical MD integration time step, which is on the order of femtoseconds. Highly coarse-grained (CG) approaches, in which more than four heavy atoms are mapped into a single CG bead, offer a way to overcome these scale limitations when the atomistic detail is not essential to the research question. Although part of the speed-up in CG simulations comes from reducing the number of particles whose motion must be integrated, the primary factor is the smoother free-energy landscape of CG systems, which allows for larger time steps and more efficient sampling (11–15).

Here, we first tested a previously developed phenomenological solvent-free four-bead CG lipid model known as the Grime lipid model (Fig. 1) (16,17), to simulate submicrometer-scale

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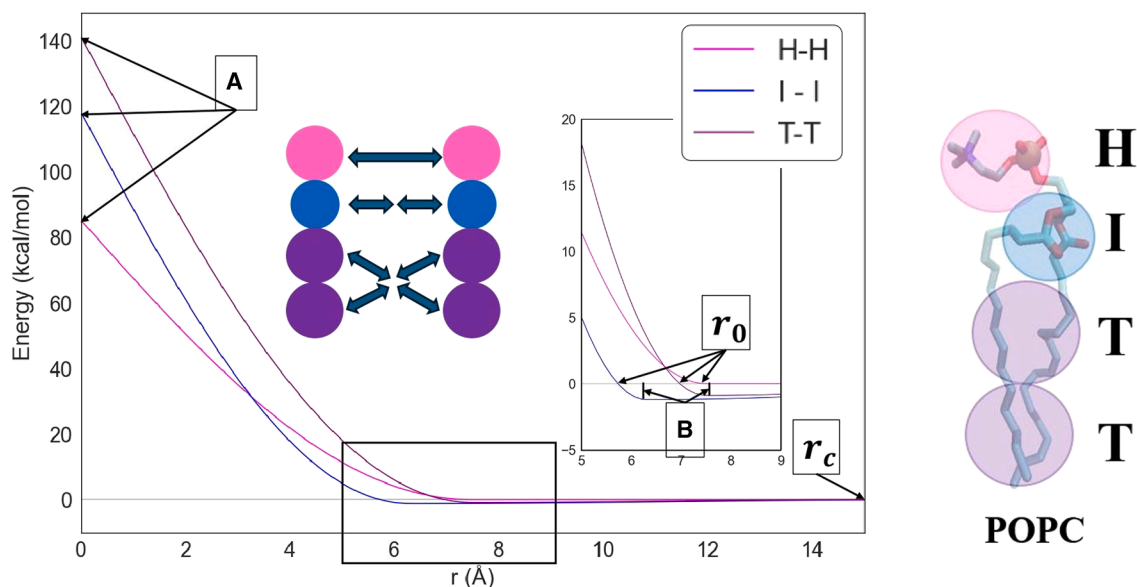


FIGURE 1 (Left) Pairwise potential of the intermolecular interaction in the CG lipid model. The head bead interaction, $H-H$, is all-repulsive ($B = 0$), whereas an attractive cohesion is in effect for the $I-I$ and the $T-T$ interaction. Cross-interactions between different bead types are all-repulsive ($B = 0$). Example parameter values for A , B , r_0 , and r_c are used for illustration. For actual parameters used in simulations, refer to the [materials and methods](#) section. (Right) Schematic representation of the four-bead Grime lipid model. Head (“H”), interface (“I”), and tail (“T”) type beads are shown as magenta, blue, and purple spheres, respectively. An approximate atomic mapping into corresponding CG beads is indicated for the prototypical lipid, 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine (POPC).

membrane patches. Although many similar CG lipid models exist (18–34), the short-ranged soft-core pairwise potential used in the Grime model has been shown to be both versatile and tunable (16). It has been widely applied in simulations of protein scaffolding (35–39), transmembrane proteins (40), lipid droplets (41–43), lipid nanocarriers (44), and viral particles (45–47). In addition, work from others on comparable models has shown capacity for elucidating raft-like membrane mixtures (48,49). However, in this study, we observed that beyond a critical membrane size, the original parameters in the four-bead Grime model for fluid bilayers became problematic. Specifically, scale-dependent issues emerged in the form of persistent membrane pores that were absent in smaller-scale simulations using the same parameters. Additionally, we observed unphysical membrane undulations that progressively distorted and destabilized the bilayer, ultimately leading to its collapse. These issues differ from artifacts caused by poor technical choices, such as overly short outer cutoffs and too infrequent neighbor list updates, which can also introduce distortions in bilayer simulations (50). We demonstrate that by optimizing the head and interface lipid bead parameters, we can effectively eliminate membrane

pore and unphysical undulations, improving the stability of large-scale bilayer simulations.

The optimized lipid parameters greatly enhance the utility of our CG model by enabling simulations with substantially larger membrane sizes. This allowed us to simulate membrane proteins in a submicrometer-scale membrane patch, which would be computationally prohibitive with atomistic or Martini force fields. PIEZO1 and PIEZO2, the primary mechanosensitive ion channels in vertebrates, are among the largest known plasma membrane proteins (>8000 total residues) (51,52). Their homotrimeric structures consist of a central ion-permeating pore region and three large transmembrane mechanosensory domains (each comprising 36 transmembrane helices), commonly referred to as “arms” or “blades.” In the absence of mechanical forces, the arms adopt a curved conformation, giving the protein an inverted dome shape with a dome surface area of approximately 700 nm^2 (53,54). This dome-shaped conformation imposes a curvature on the membrane far beyond the channel, known as the PIEZO’s membrane footprint (55).

To date, PIEZO’s membrane footprint has been explored in MD simulations to limited spatial extents using membrane sizes of 800 to 2500 nm^2 , in all-atom and Martini models (56–59). We previously

showed that small membrane patches tend to flatten at the boundary, thereby altering the channel's curved resting conformation (56). Here, using a membrane patch of 14,000 nm², we investigate the footprint shape of PIEZO under various membrane conditions without boundary artifact.

In summary, we demonstrate that our optimized CG lipid models are effective in capturing the structural and dynamic interplay between proteins and membranes. Our methodological insights could contribute in addressing scale-dependent limitations in similar CG approaches. Furthermore, our refined CG model serves as a timely tool for investigating long-range dynamic membrane deformations induced by complex topologies of membrane-embedded proteins, an area that remains challenging to explore using existing continuum models or (near-)atomistic MD simulations.

MATERIALS AND METHODS

CG computational models

Lipid model: Four-bead Grime lipid

The potential energy function of the Grime CG lipid model is given by the following (16).

$$U(\mathbf{r}) = \sum_i U_{\text{bond}}(r_{i,i+1}) + \sum_i U_{\text{angle}}(\theta_{i,i+1,i+2}) + \sum_{i>j} U_{\text{pair}}(r_{ij}) \quad (1)$$

with

$$\begin{aligned} U_{\text{bond}}(r) &= K_b(r - r_0)^2 \\ U_{\text{angle}}(\theta) &= K_\theta(\theta - \theta_0)^2 \end{aligned} \quad (2)$$

$$U_{\text{pair}}(r) = \begin{cases} -\frac{A \cdot 2r_0}{\pi} \sin\left(\frac{\pi}{2r_0} r\right), & \text{if } r \leq r_0 \\ -\frac{B \cdot (r_c - r_0)}{\pi} \sin\left(\frac{\pi}{2} + \frac{\pi}{r_c - r_0} (r - r_0)\right), & \text{if } r_0 \leq r \leq r_c, \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where $K_b/(k_B T \text{ \AA}^{-2}) = 25$ is the bond force constant, $K_\theta/k_B T \cdot \text{rad}^{-2}$ is the angle potential parameter that controls the membrane bending rigidity, $r_0/\text{\AA}$ is the bead size, and $r_c/\text{\AA}$ is the interaction cutoff distance. In the four-bead variant, each lipid molecule consists of a head bead ("H"), an interface bead ("I"), and two tail beads ("T") (Fig. 1). The mass of each CG bead was set to be equal by evenly distributing the mass of one 1-palmitoyl-2-oleoyl-glycero-3-phosphocholine (POPC) lipid over the four beads. The model reproduces the experimental area per lipid (APL) for POPC ($64.3 \pm 1 \text{ \AA}^2$) (60), which is also consistent with all-atom MD simulations (e.g., using the CHARMM36 force field (61)) (62,63), at the correct bending rigidity ($\sim 18\text{--}25 k_B T$) (64).

Using the originally proposed set of parameters ($A = 50 k_B T$, $B = 0.311 k_B T$, $B_{\text{cross-terms}} = 0$, $r_0(H-H, I-I, T-T) = \{0.75, 1, 1\} \ell$, $r_c = 2r_0$ with $\ell = 7.5 \text{ \AA}$) (16), the resulting bilayer membrane for a system of 20k lipids can be controlled via a single parameter, the angle potential constant of the molecular lipid, K_θ , to yield a

bending stiffness ranging from a few to $\sim 100 k_B T$. In the optimized set of parameters resulting from this study, we use bead sizes of $r_0(H-H, I-I, T-T) = \{1, 0.84, 1\} \ell$ and an increased range of the I - I cohesion by 10%, $r_c(I-I) = 1.1 \cdot r_c$.

Protein model: PIEZO ENM

The PIEZO protein is modeled using an elastic network model (ENM), where harmonic bonds connect the CG beads. The ENM protein model is effective while being very simple, accurately capturing Gaussian fluctuations (65–69). Many ENM-based approaches have been successfully applied in protein modeling (70–79). The full-length PIEZO2 cryo-EM structure (PDB: 6KG7) serves as the template for constructing our protein model. To match the lipid resolution, the PIEZO model has an average resolution of approximately three amino acids per CG bead.

Each CG bead is positioned at the center of mass of the Ca atoms from every three consecutive amino acid residues. A few exceptions exist where a bead represents four residues, primarily in specific loop regions extended outside the transmembrane region. This grouping is necessary to account for an extra residue at the end of a loop outside of the transmembrane region of the protein. Since these residues are expected to have minimal interaction with the membrane, representing them using an additional bead would be less accurate and unnecessarily increase computational cost without significant gain in accuracy. Therefore, the center of mass for these terminal loops may be shifted to incorporate the loops with one or two extra residues, which avoids introducing an extra bead.

At this level of coarse-graining, we do not distinguish between the amino acid sequences of PIEZO1 and PIEZO2. Instead, we refer to our model as a general PIEZO representation, designed to investigate the interplay between protein shapes and membrane properties. The beads are typed so as to match the character of the bilayer membrane (traversing the membrane, the bilayer has the types H-I-T-T-T-T-I-H) (Fig. 2 a). PIEZO beads outside of membrane (e.g., loops or the "Cap" domain and the "beam") are typed as all-repulsive head bead types. For symmetric transmembrane proteins such as homotrimeric PIEZO, it is advantageous to explicitly ensure that CG mapping satisfies structural symmetries (80–82). Although generalizing and extending CG models for membrane proteins presents promising opportunities (68,69,83–85), this is beyond the scope of the present study.

In the ENM, the CG protein beads are connected by harmonic bonds. All harmonic bonds were assumed to have the same force constants, and this value was determined to be $2 \text{ (kcal/mol)/\AA}$ by comparing root-mean-squared fluctuations (RMSFs) to a reference all-atom MD simulation (Fig. 2 b). The equilibrium distances for the harmonic bonds were binned using bins of $0.25 \text{ \AA}$ to reduce the required number of bond types. To find the optimal cutoff distance between protein beads, we scanned a range of values, beginning with $13.25 \text{ \AA}$, the distance needed to maintain the structural integrity of the PIEZO repeat A in the membrane. The scan was performed by incrementally adding $0.25 \text{ \AA}$ until the optimal cutoff of $16.25 \text{ \AA}$ was reached. Increasing the cutoff further would be counterproductive, as it would both make the protein more rigid and require more computing resources.

RMSF analysis was performed to compare the dynamics of the all-atom and CG elastic network models of the PIEZO protein. Both models were simulated in a vacuum to ensure a direct and unbiased comparison. This approach was necessary to prevent double counting of membrane-induced effects. Had the CG ENM model been based on a membrane-embedded structure, the membrane's constraining effect on the protein's fluctuations would have been implicitly included twice, potentially skewing the results

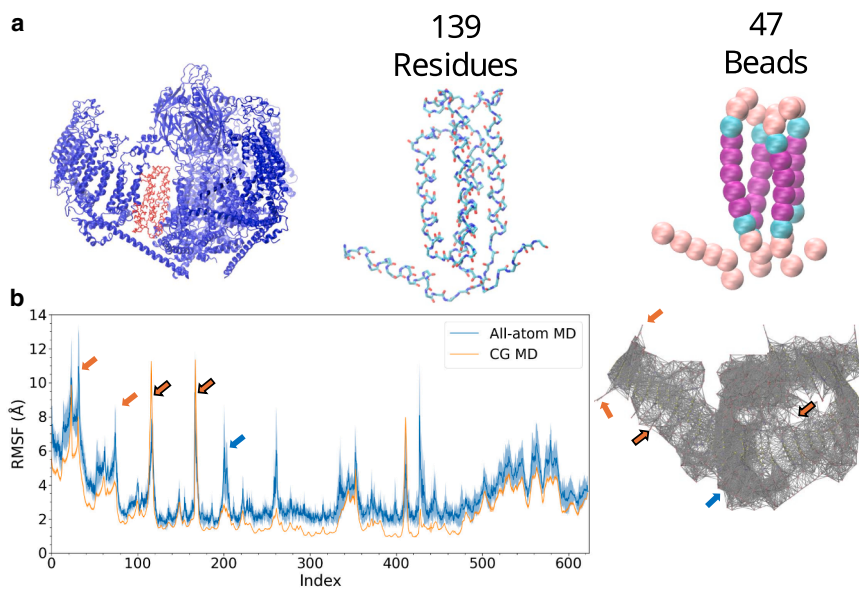


FIGURE 2 Coarse-grained model of PIEZO channel. (a) Coarse-graining the structure of PIEZO. The cryogenic electron microscopy structure of full-length PIEZO2 (PDB: 6KG7) (54) is shown as ribbon diagram in the left panel (blue, with the 139-residue PIEZO repeat A unit in red). Middle panel shows the backbone of repeat A unit as sticks. Right panel shows the corresponding CG-mapped repeat A unit consisting of 47 beads as spheres (bead coloring is consistent with Fig. 1). (b) Root mean-square fluctuation (RMSF) of protein backbone computed from simulations of all-atom and CG PIEZO. The RMSF is calculated and averaged to one arm, with the shaded areas indicate standard deviation across the three arms, and the CG bead numbering (623 beads per arm with a total of 1869 beads per PIEZO) is used in this graph instead of atom numbering. The elastic bonds connecting the CG beads in PIEZO are visualized in the panel on the right. The arrows indicate that the positions of mismatched peaks are due to highly flexible loop regions.

and hindering an accurate representation of the models' intrinsic dynamics. The exact CG mapping employed is available in our GitHub repository.

Simulation procedures

Flat bilayer system setup

Flat bilayers were constructed by placing lipids on an ideal square X-Y grid defined by the average positions of the equilibrated APL. More details can be found in our previous work (16).

CG simulations

All simulations were performed with LAMMPS (version Mar 24, 2022) (86,87), using the Nosé-Hoover extended Lagrangian formalism for controlling pressure and temperature (88–90). The integration time step used in all simulations is $\tau = 50fs$. The weak coupling times of $100,000\tau$ were used to minimize perturbations in the evolution of the system. The simulations were carried out either in lateral-stress-less *NPT* ($T = 300K$, $P_{xy} = 0 atm$) or *NVT* ($T = 300K$) ensembles as appropriate.

All-atom simulation

A reference atomic simulation of the PIEZO2 structure resolved by cryogenic electron microscopy (PDB: 6KG7) (54) was prepared using CHARMM-GUI tools (91–95). The CHARMM36m force field (6) was used to describe the protein. This all-atom simulation was also performed with LAMMPS (version Mar 24, 2022) using a time step of $1fs$ (86,87). Lennard Jones and coulombic interactions were calculated using the built-in CHARMM force field features in LAMMPS with a dual-cutoff scheme of $20 \text{ \AA}$ (inner) and $22 \text{ \AA}$ (outer), respectively. Initial velocities were set corresponding to $T = 303.15 K$, and maintained with a Nosé-Hoover thermostat with a damping constant of $\tau_{\text{damp}} = 100$ at constant volume (*NVT* ensemble). After a steepest-descent energy minimization for 5000 steps, the simulation was run for $1 ns$, in vacuo for the purpose of computing RMSF. Under these conditions, the simulation temperature was kept at an average of $303.188 \pm 1.67K$.

Embedding PIEZO into the membrane

The membrane-embedded PIEZO system was first prepared using a small membrane patch of 45k lipids ($\sim 30 \times 30 nm^2$). PIEZO was placed in the center of the incision and overlapping lipids (within $\ell = 7.5\text{\AA}$) were removed. The system was then subjected to energy minimization and equilibration. During equilibration, the lipids progressively fill any gaps surrounding the protein. Any ejected lipids were manually removed, after which the simulation was resumed for final equilibration. The final membrane-embedded PIEZO structure was saved as a PDB file, serving as template for embedding PIEZO into larger membrane patches.

Simulation single PIEZO in a lipid bilayer

The membrane-embedded PIEZO patch was placed in the desired membrane architectures, and the overlapping lipids (within $\ell = 7.5\text{\AA}$) were removed. Depending on the simulation, PIEZO was frozen in place or fixed at the location of a central bead allowing protein fluctuation and free rotation. In LAMMPS, these were achieved by invoking “fix spring tether or fix spring/self.” The bending rigidity is effectively controlled by the lipid angle potential parameter, K_{ϕ} . For simulations under externally applied tension, the box size is altered (in *NVT*) to increase APL and stretch the membrane by invoking “fix deform.” All simulations were conducted for at least 4 million integration time steps, τ . In systems requiring further equilibration (such as those in which the box size was altered), the first 2 million τ were discarded from the analyses.

Analysis of trajectories

We used in-house Python scripts relying on packages MDAnalysis (96,97), numpy (98), and pandas (99) to perform all analyses. Visualization and plotting were performed using matplotlib (100), seaborn (101), Microsoft Excel (version 16.84) (102), VMD (version 1.9.4) (103), and OVITO (version 3.10.3) (104). APL was estimated as the time-averaged X-Y projected APL based on the equilibrated box dimensions and averaged over three independent replicas. Membrane bending rigidity, k_c , was determined using the fluctuation spectrum method (105).

Lateral pressure

The lateral pressure components along X and Y, P_{xx} and P_{yy} , respectively, were obtained directly from the stress tensor in LAMMPS at every 5000 τ . The X-Y averaged instantaneous lateral pressure is given by $(P_{xx}+P_{yy})/2$. A rolling average procedure was used to de-noise the pressure time series. The lateral pressure of the membrane indicates the lateral stress of the membrane with $p = 0$ corresponding to the desirable “stressless” state in the CG ensemble. We remark that the interpretation of pressure in CG simulations is nontrivial (13) and further that the usual equation for surface tension ($\gamma = L_z/n \times (P_{zz} - (P_{xx}+P_{yy})/2)$, where L_z is the Z box dimension, and n is the number of surfaces (106,107)), may not always strictly be interpreted as the membrane’s lateral pressure, especially considering the significant curvatures manifested in the PIEZO dome and footprint.

Lateral pressure profiles

Simulations of various sizes (containing 128, 512, 1152, 2048, 4608, and 8192 lipids) were conducted to investigate the lateral pressure profile and how it depends on the lipid parameters. These simulations were run for 200,000 τ in the NPT ensemble for equilibration, and then an additional 1 million τ for data collection in the NVT ensemble. The lateral stress profile is computed using the method of planes as implemented in LAMMPS (“compute stress/mop/profile”) (108,109). This method is preferred over binning atomic stresses when simulating heterogeneous systems and interfaces (108).

PIEZO membrane footprint

The PIEZO footprint was calculated by averaging the position of membrane interfacial beads in 2D (X-Y planar) 5Å radial bins from PIEZO’s center of mass. For each bin, the Z positions of the lipid interfacial beads were averaged.

RESULTS AND DISCUSSION

Bilayer membranes formed by the four- and five-bead lipid models are attractive for various reasons. First, they (like other resolutions) can have their membrane bending stiffness controlled by a single parameter, typically the angle potential parameter for the lipid molecule, to form a fluid-phase bilayer membrane with mechanical bending stiffness in the range of a few to $\sim 100 k_B T$. Furthermore, the four- and five-bead resolutions are intuitively appealing due to their ability to semiquantitatively reproduce the correct lateral pressure profile with a high pressure peak in the interfacial region of each membrane leaflet (20). This feature is especially important for studying integral membrane proteins that span the membrane, with PIEZO being even more critical due to its sensitivity to lateral stress. The four-bead model is preferred over the five-bead model primarily because fewer beads are needed to describe the lipid molecule, making the simulation more computationally efficient. Another tangible benefit is that since the four-bead membrane is less thick than the five-bead model, it is more responsive to a given change in

angular potential parameter, K_θ , which becomes increasingly important as the size of the membrane increases. Recall that for a given parameter set, the resulting bilayer membrane gets much less stiff when the size of the system is increased (an effect that is due to the pinning at the box boundary). So, the angle parameter will have to be tightened to simulate a bilayer membrane at similar bending stiffness when the box size increases (16). Alternatively, the range of the attractive cohesion interaction may be increased (110).

The four- or five-bead lipid models are commonly used to simulate membrane patches of up to thousands of nanometers square in size. In this study, we aim to test the stability of our lipid model for simulating large membrane proteins in a submicrometer membrane patch. What is rather unexpected is that membrane poration, which is not observed in smaller membranes, appears above a certain critical size for a given parameter set. Furthermore, when the size of the box increases, the membrane bilayers can have a tendency to undergo unphysical distortions by curling up and crumbling, depending on the specific parameters used. Therefore, in the current work, we first needed to optimize the lipid parameters to obtain a stable, planar membrane patch of adequate size. Using this refined model, we then examined PIEZO’s membrane footprint in flat membranes under varying membrane bending rigidities and tension conditions.

Resolving poration and instability issues in simulations of large membrane patches

Using the originally proposed parameters for the four-bead lipid model, we can readily realize fluid bilayer membranes with the properties previously reported (16). However, increasing the system size from a simulation box containing 20k to 45k lipids unexpectedly leads to the formation of persistent pores in the membrane (Fig. 3). For this specific case, visual analysis of the simulation frames revealed the diffusion of persistent pores within the lipid bilayer. We note that in the original study (16), this undesirable artifact was not observed in any of the tested morphologies as the system sizes were less than or equal to 20k lipids. As these pore structures were deemed undesirable artifacts rather than physically relevant features, no quantitative analysis was performed.

As previously suggested (111,112) and demonstrated for the Grime lipid (16), the major characteristic of a lipid in determining the assembly morphology is the molecular packing ratio, or how “conical” the molecule is. This immediately suggests that decreasing the size of the head bead in the lipid model should help resolve the poration issue. Indeed,

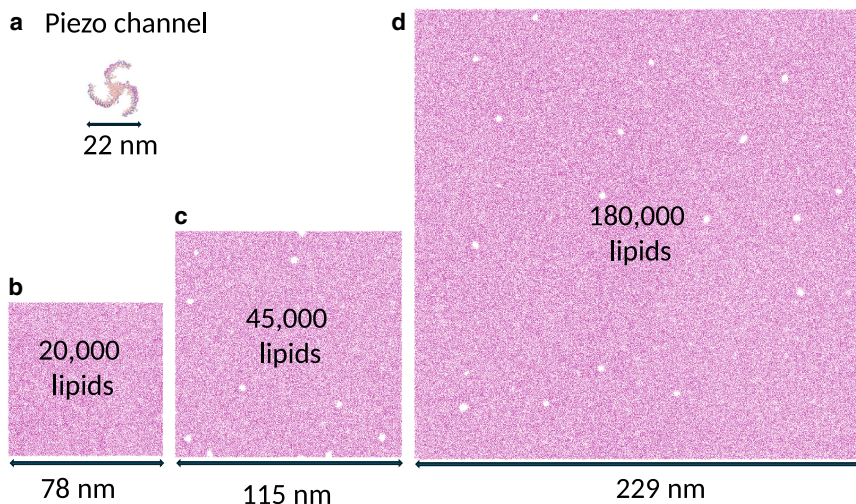


FIGURE 3 Size-dependent poration in membrane patches. (a) The size of the PIEZO protein is indicated for comparison. The three significant membrane sizes simulated are (b) 20k lipids ($X = Y \approx 78.5$ nm), (c) 45k lipids ($X = Y \approx 115.3$ nm), and (d) 180k lipids ($X = Y \approx 228.8$ nm). The poration issue can be seen above the critical size for the reference set of parameters at around or above 45k lipids, whereas it is completely absent for the system of 20k lipids.

at any membrane size tested, poration is completely removed by decreasing the size of the lipid head bead from $R_{HH} = 0.75\ell$ to $R_{HH} \leq 0.70\ell$, where $\ell = 7.5\text{\AA}$ is the characteristic length scale in the simulation (Fig. 1). Fig. S1 demonstrates the result of tuning with various R_{HH} and K_θ in a membrane size of 45k lipids (Fig. 3 c), visualizing the resulting changes in APL and stability.

However, upon resolving the issue with operation using $R_{HH} \leq 0.70\ell$, we encountered an instability issue when simulating a large membrane patch in *NPT* ensemble (Fig. 4 a). Analysis of previous data showed that the poration and instability issues emerged using originally proposed model parameters and were not introduced by lowering the R_{HH} from 0.75ℓ to 0.70ℓ . A pragmatic approach for this would involve simulating instead the corresponding canonical (*NVT*) ensemble at an appropriate box size to test if each issue can be addressed individually. We note that simulating in the *NVT* as opposed to the *NPT* ensemble usually is not a strong limitation since sampling in the two ensembles would be consistent for large enough systems (113,114). However, for bigger systems (e.g., 180k lipids) we tend to see that there is no “good” choice of box size for *NVT* ensemble in the sense that the lateral-stress free state ($\langle P_{xx}, P_{yy} \rangle \approx (0,0)$) cannot be achieved (Fig. 4 b). This observation may seem baffling at first. The negative sign of the lateral pressure suggests that the membrane favors contraction, and one might think that the box is simply too large and resizing for a smaller box can resolve the issue. However, this is not the case: decreasing the box size (i.e., decreasing APL) will increase the amplitude of the unphysical sinusoidal distortion without ever reaching the state of vanishing lateral stress. Note that these simulations

were performed in the canonical (*NVT*) ensemble so that the box size could be precisely controlled. In all cases for the large membrane, increasing the box size inevitably leads to increasing (negative) lateral pressure as expected, but decreasing the box size leads to ever-larger amplitude sinusoidal distortions and the persistence of (negative) nonzero lateral stress.

Several other attempts at tweaking the poration-fixed model ($R_{HH} = 0.7\ell$), including modifying in addition the angle potential parameter (K_θ) or the strength (Fig. 1; Eq. (3), parameter value B) or range of the cohesion (Fig. 1; Eq. (3), parameter value r_c), in combination with the head bead-size modification that eliminated poration, were all unsuccessful. A solution that did prove effective was to solve the poration issue by changing the lipid molecule shape, not by resizing the head bead, but both the head and interface beads. When this is done in conjunction with a modification to the interface-bead cohesion (increasing the range slightly to facilitate stability), the resulting model will produce a stable fluid bilayer membrane that can be simulated without pores, is lateral-stress free, and remains stable over 10 million τ and up to the largest membrane size tested ($0.05 \mu\text{m}^2$) (Fig. 4 c). Fig. S2 demonstrates the result of tuning with various R_{HH} , visualizing the resulting changes in APL and stability. Upon elucidating the lateral stress profile of the membrane for both the original and optimized parameters, it is seen that both profiles exhibit similar overall structure but that the optimized parameters include a characteristic “shoulder” feature near the bilayer interface (Fig. 4 d). We interpret this as indicating a redistribution of the membrane lateral stresses, alleviating both the poration issue and the unphysical distortions by increasing

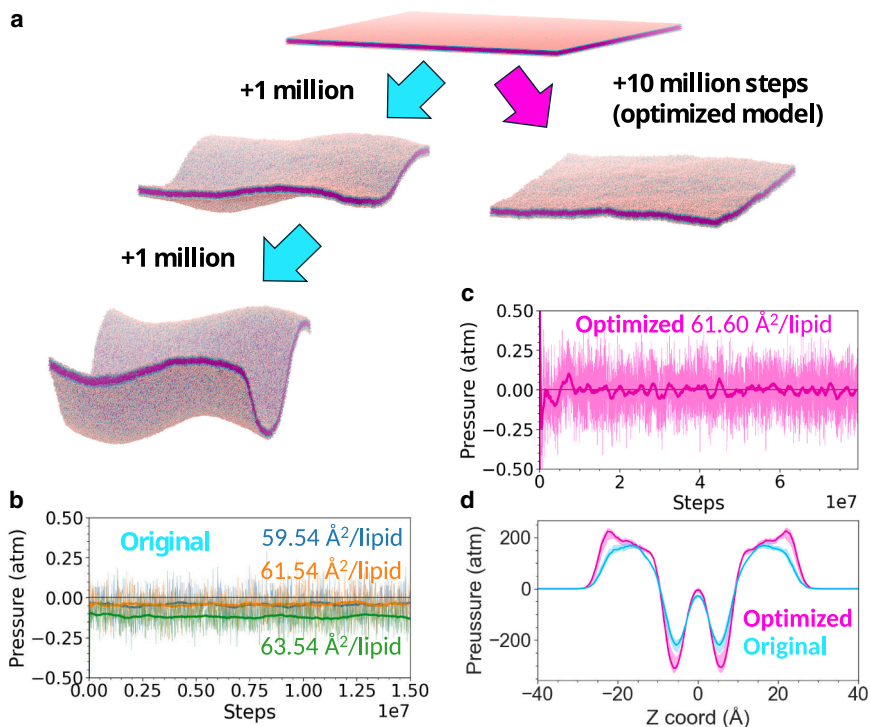


FIGURE 4 Stability and lateral tension of the original and optimized lipid model. (a) Illustration of the destabilization of membrane caused by residual tension in Grime lipid model with its original parameters (cyan) compared the optimized model with adjustable membrane tension (magenta) currently at zero tension. (b) Persistence of non-zero tension with the original model when run with different APLs, {59.54,61.54,63.54}Å². (c) Stable trend of no tension with the optimized model at APL = 61.60Å². (d) Illustrates the lateral pressure profile of the Grime lipid model using both originally and optimized parameters. The features of this lateral pressure profile depend on system size, with larger boxes exhibiting increasingly averaged characteristics. Results shown here come from a 2 × 2 lipid bilayer patch, totaling eight lipids, where local features are clearly distinguishable. The shaded areas indicate the minmax range of values over 5 independent runs.

membrane stability in a physically relevant stress distribution. Importantly, it is possible to simulate the membrane without residual lateral stress (in NVT) for the employed range of bending stiffness, $k_C = (18 - 34)k_B T$, even at constant APL (Fig. S3).

To further elucidate this phenomenon in absence of periodic boundary condition, a planar disk-shaped membrane bilayer was simulated in vacuo. This “pizza” configuration was observed to distort and spontaneously curl up into a vesicle when using the original parameters (Fig. S4). However, with the optimized parameters, the membrane remained planar despite having the same bending rigidity, k_C . It is important to remark that the nonvanishing lateral stress is not the only contributor to this behavior; a reduction in the membrane edge line tension also plays a role, but their relative contributions are nontrivial to disentangle.

The optimized model enables us to conduct simulations with vanishing lateral stress, $\langle P_{xx}, P_{yy} \rangle \approx (0,0)$. However, this does not always make it practical to employ standard barostat algorithms due to the solvent-free nature of the CG model and the increasing amplitude of height fluctuations in large systems. It is evident that a larger magnitude of undulation reduces the total horizontal membrane area in the XY plane. Fig. S5 shows that the APL (measured as the number of lipids divided by the horizontal membrane area) decreases with system size roughly logarithmi-

cally, consistent with theoretical predictions (115). For this reason, we suggest a pragmatic solution whereby the canonical (NVT) ensemble is employed, and the initial guess of the box size of the large membrane is estimated from extrapolation of the APL of smaller membrane patches (Fig. S5) and followed by fine-tuning this value to achieve vanishing lateral stress.

Simulation of PIEZO footprint in planar membranes

Using our optimized CG lipid model, we can now simulate a stable membrane patch of submicrometer size, large enough to accommodate multiple PIEZO channels without their footprints overlapping. In this proof-of-concept study, we systematically investigated the effect of the bilayer mechanical properties on the membrane footprint of a single PIEZO channel. Our PIEZO model is based on the cryo-EM structure of full-length PIEZO2 solved in detergent (54), which has an inverted dome shape with an axial height of 9 nm and a projected area of 450 nm² (54). Once embedded in the lipid bilayer, the size of the membrane footprint is predicted to depend on the shape of the protein, the membrane bending rigidity (k_C), and membrane tension (γ) (55,116).

The PIEZO protein is modeled as a simple ENM, meaning that the protein conformation is restrained to fluctuate around its initial (closed) state.

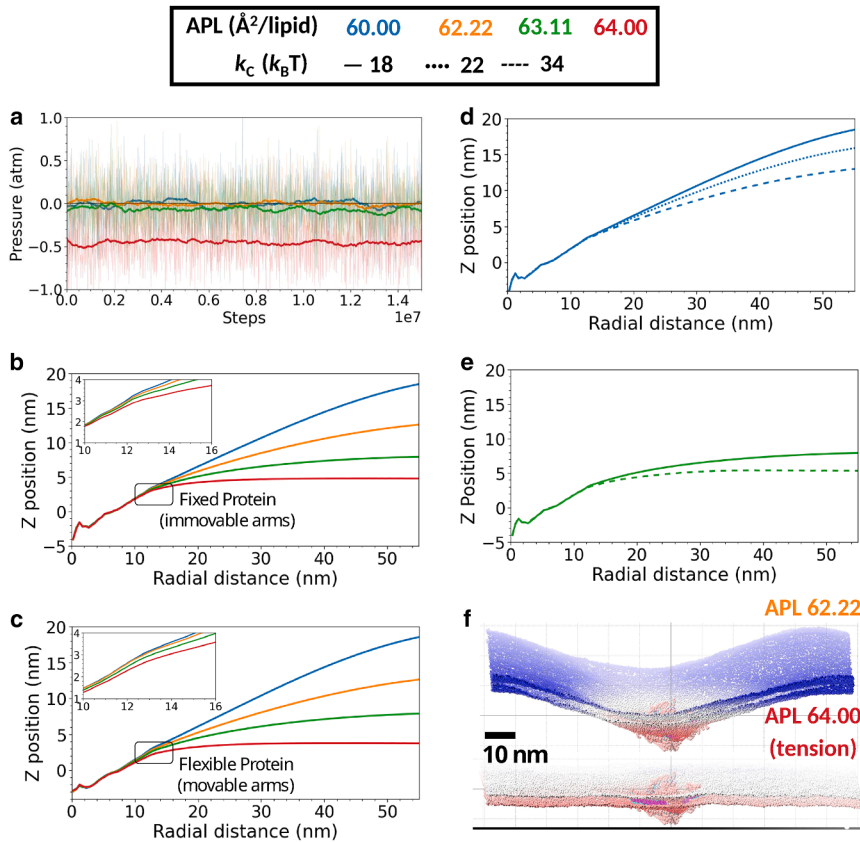


FIGURE 5 PIEZO's membrane footprint plotted as average membrane Z-position as a function of radial distance to PIEZO's center. The simulation box has 45k lipids. (a) Lateral pressure at different APL values. PIEZO's footprint at varying lateral pressure with the protein conformation fixed (b) or flexible (c). Axis units of inserts are identical to those used in the main plots. PIEZO's footprint with increasing bending stiffness, $k_C = \{18, 22, 34\} k_B T$ at zero tension (d) or at low tension (e). (f) Illustration of the PIEZO footprint corresponding to (C) with the membrane colored by Z-coordinates (along membrane normal) in the absence ($60.00 \text{ \AA}^2/\text{lipid}$) or presence of lateral stress ($64.00 \text{ \AA}^2/\text{lipid}$).

Therefore, this ENM is not suitable for studying the closed-to-open transition, as accurate modeling of conformational transition would require more complex models, such as the ones with multiple-basin potentials. Here, we focus on investigating membrane deformation induced by the closed PIEZO protein. Our previous all-atom MD simulations of the PIEZO1 channel in an 800-nm^2 membrane patch have shown that periodic boundary condition (PBC) can flatten the membrane at the simulation box boundary, thereby destabilizing the closed state of PIEZO1 (56). By using a highly CG model here, we can simulate an exceptionally large membrane patch of $14,000 \text{ nm}^2$, significantly larger than the protein itself (approximately 20 nm in diameter)—which helps to minimize PBC artifacts on the membrane footprint size.

We first investigated the PIEZO footprint under varying membrane tension. Keeping the protein conformation fixed and physiological membrane bending rigidity at $k_C = 18 k_B T$, we introduced lateral stress by increasing the size of the simulation box in each independent simulation (Fig. 5 a). As predicted by continuum elastic theory (55), the size of the PIEZO footprint decreased as the membrane was stretched (i.e., as the membrane tension increased) (Fig. 5 b).

To test whether our elastic network PIEZO model responds to membrane tension, we conducted the same set of simulations without fixing the protein conformation. We observed that the PIEZO dome remains largely unchanged at low resting tension. However, at a lateral pressure of -0.5 atm , the curvature of the dome began to decrease due to flattening of the PIEZO arms, consistent with the membrane tension sensing mechanism of the PIEZO channels (Fig. 5 c).

To investigate the effect of membrane rigidity, we simulated fixed PIEZO in three tensionless bilayers with bending rigidity that spans a physiologically relevant range $k_C = 18, 22$, and $34 k_B T$ (Fig. 5 d). We observed that the membrane flattens asymptotically up to the boundary of the simulated bilayer. This observation is consistent with predictions from continuum elastic theory, which states that the footprint size is theoretically infinite under zero tension (55). However, whereas the continuum elastic model predicted that the decay length of the footprint λ increases with the rigidity of the membrane ($\lambda = \sqrt{k_C/\gamma}$) (55), we found that the footprint decays faster, hence smaller λ , as the membrane becomes more rigid in our simulations.

Because the footprint decay length is infinite under zero tension, we cannot entirely rule out the PBC

artifact in PIEZO's footprint, even in the largest planar membrane simulations. In living cells, however, the plasma membrane has nonzero resting tension of $\gamma = 0.01\text{--}1.0\text{ }k_B T/\text{nm}^2$ (117,118). Thus, we simulated the PIEZO footprint at a membrane APL of $63.11\text{ }\text{\AA}^2$ (Fig. 5 b, green). Since this system represents PIEZO footprint size under low resting tension with negligible PBC artifact, we further simulated the PIEZO footprint under membrane bending rigidity of 18 and $34\text{ }k_B T$ (Fig. 5 e). We again observed the same trend as in Fig. 5 d, where the footprint decays more rapidly as the membrane becomes more rigid. One difference between the continuum results and our CG simulation is that in continuum theory, the bending stiffness is a global parameter that can be manipulated without affecting other membrane properties, whereas in MD simulation, it emerges from the rigidity of individual lipids, mimicking how bending rigidity in plasma membrane varies with lipid composition. A change in the lipid angle parameter will affect APL and bending stiffness, thus coupling lipid rigidity, bending modulus, and lateral packing. We suspect that the footprint decay can be a consequence of this interplay. Further study is needed to enable direct quantitative comparison with continuum theory results and available experimental data.

CONCLUSIONS AND PERSPECTIVES

At the micron scale, the cell membrane and its embedded proteins display amazing complexity in their shape and organization. To explore the underlying physical principle driving such phenomena, there is a need to simulate the equilibrium ensemble and dynamical behavior of protein diffusion and long-range protein-membrane and protein-protein interactions. The CG model employed here is designed to fill the gap between the elastic continuum model of membrane deformation and the atomistic or near-atomistic MD approaches.

We first demonstrated how to systematically resolve scale-dependent issues arising only in membrane systems above a certain critical size, including poration and unphysical sinusoidal distortions leading to crumbling. Although this was done for a specific CG lipid model, the four-bead Grime lipid, we anticipate that the insights gained will be useful for resolving similar problems in related CG methodologies.

Using our optimized CG model, we demonstrate that increasing membrane bending rigidity or lateral tension reduces PIEZO's membrane footprint. These observations may have far-reaching implications for membrane curvature-mediated protein-protein interactions. Previous CG models of protein-induced mem-

brane curvature have primarily focused on proteins bound to membrane surfaces. In contrast, our CG protein and lipid models are designed to simulate proteins fully embedded in the membrane and tailored to study long-range mechanical coupling between proteins and lipids.

Although further parameterization, ideally in conjunction with atomistic simulations, is needed to refine the extent of CG protein conformational changes, our model effectively captures the response of the PIEZO dome and footprint to membrane mechanical properties without the previously reported periodic boundary effect that artificially flattens the PIEZO footprint.

The optimized lipid parameters greatly enhance the utility of our CG model by enabling substantial extension of membrane size. This will, in turn, allow us to simulate multiple PIEZO channels in a submicrometer-scale membrane patch. For instance, a system of five PIEZO channels embedded in a $350 \times 350\text{ nm}^2$ membrane corresponds to a channel density of approximately 40 channels per μm^2 . Such multi-channel simulations may provide valuable insights into channel clustering or interchannel interactions arising from overlapping membrane footprints at physiologically-relevant channel densities. Hence, generalizing and extending our CG models will open promising avenues for studying large-scale membrane-protein systems.

DATA AND CODE AVAILABILITY

Lipid parameters (original and optimized), the CG mapping scripts, final CG PIEZO model, LAMMPS CG-MD input files, and all analysis scripts are available on GitHub at <https://github.com/charliewen95/LAMMPS-CG-membrane.git>.

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AUTHOR CONTRIBUTIONS

Y.L.L. and J.J.M. contributed to conceptualization and design of the study. C.Y.W. conducted the experiments, collected the data, and analyzed the data. All authors contributed to interpretation of the

results, edited and revised the manuscript, and approved the final version for publication.

DECLARATION OF INTERESTS

The authors declare no competing interests.

SUPPORTING MATERIAL

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